CHE 201 Fall 2019 HW 11

Bader Hasan

TOTAL POINTS

100 / 100

QUESTION 1

1 1 **15 / 15**

✓ **- 0 pts** Correct

- **- 5 pts** u1,u2 incorrect, so Q also incorrect
- **- 5 pts** S1 , s2 incorrect
- **- 1 pts** delta S unit incorrect
- **- 3 pts** delta S answer missing
- **- 1 pts** delta Q is negative in this case
- **- 8 pts** delta S equation for idea gas was incorrect

QUESTION 2

2 2 **15 / 15**

✓ **- 0 pts** Correct

- **- 2 pts** Delta S is positive

QUESTION 3

3 3 **15 / 15**

✓ **- 0 pts** Correct

- **- 7.5 pts** W = ?
- **- 2 pts** Work is negative
- **- 2 pts** Q is negative
- **- 3.5 pts** Wrong value for s1. s1 = 1.9690 Btu/(lb*R)
- **- 2 pts** Wrong conversion from oF to oR

QUESTION 4

4 4 **15 / 15**

✓ **- 0 pts** Correct

- **- 5 pts** Th = 336 K (63 oC)
- **- 5 pts** CoP = Tc/(Th-Tc)
- **- 15 pts** No answer

QUESTION 5

5 5 **20 / 20**

✓ **- 0 pts** Correct

- **- 2 pts** c) the rate of entropy production is positive
- **- 3 pts** a) Welectric = 2,990 kW. Q = -158 kW.
- **- 6 pts** c) rate of entropy production = -Q/To = 0.53 kW/K

QUESTION 6

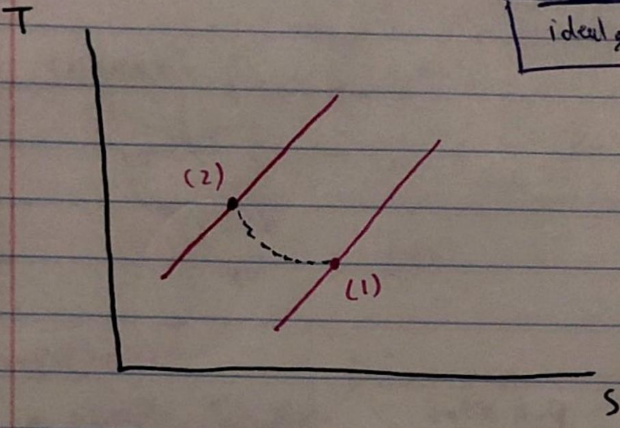
6 6 **20 / 20**

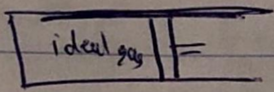
✓ **- 0 pts** correct

- **- 5 pts** (1) definition of delta Sp; (2) why delta Sp =0; (3) why s1f, s1i, s2f, s2i were not included for the following calclation?
- **- 3 pts** Its 400K based on the given question
- **- 1 pts** unit of K not degree K
- **- 5 pts** The delta S equation for idea gas was incorrect
- **- 6 pts** final P =?
- **- 2 pts** entropy produce calculation incorrect, P =400 noT 300K
- **- 6 pts** entropy produce=?
- **- 20 pts** no submission
- **- 5 pts** delta S calculation incorrect
- **- 1 pts** Unit for T2
- **- 3 pts** calculation of m1 & m2=? Final P incorrect
- **- 3 pts** (1) why the delta piston was mentioned at the beginning and was not included in the following calculation? need the explanation for each step if it was neglected

Bader Hasan

1)





given

$P_1 = 150\text{ kPa}, \quad P_2 = 500\text{ kPa}$

$T_1 = 300\text{ k}, \quad T_2 = 370\text{ k}$

$W = -300\text{ kJ}$

energy balance

$$\Delta U + \cancel{\Delta KE}^{0} + \cancel{\Delta PE}^{0} = Q - W$$

$$\Delta U = Q - W \iff Q = \Delta u + W$$

a) $\Delta U = n[\bar{u}(T_2) - \bar{u}(T_1)]$ change in internal energy

$Q = n[\underbrace{\bar{u}(T_2) - \bar{u}(T_1)}_{\text{Table 23}}] + W$ Total heat transfer

$n = 0.1\text{ kmol}, \quad \bar{u}(T_1) = 6229 \frac{kJ}{kmol}, \quad \bar{u}(T_2) = 7689 \frac{kJ}{kmol}$

$Q = 0.1[7689 - 6229] + (-300)$

$Q = \boxed{-154\text{ kJ}}$

b) $\Delta S = n[\bar{s}(T_2) - \bar{s}(T_1) - \bar{R}\ln\frac{P_2}{P_1}]$

$T_2 = 370\text{ k}$

$\bar{s}(T_2) = 203.842 \frac{kJ}{kmol.k}$ At $T_1 = 300\text{ k}$

$\bar{s}(T_1) = 197.7 \frac{kJ}{kmol.k}$

$\Delta s = (0.1)[203.84 - 197.7 - 8.314\ln(\frac{500}{150})]$

$$\boxed{\Delta S = -0.389 \frac{kJ}{k}}$$

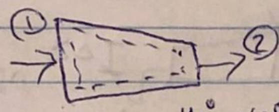
1 1 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** u1,u2 incorrect, so Q also incorrect
- **- 5 pts** S1 , s2 incorrect
- **- 1 pts** delta S unit incorrect
- **- 3 pts** delta S answer missing
- **- 1 pts** delta Q is negative in this case
- **- 8 pts** delta S equation for idea gas was incorrect

②

②

scheme: 1 MPa, 200°C



40°C, x = 83%

40°c, x = 83%

Engineering model

$\Delta KE = \Delta PE = 0$

steady state

$Q = 0$

one inlet/outlet

energy balance

$\dot m(\Delta h) = \cancel{\dot Q}^{0} - \dot W$

$\dot m \Delta h = +W$

h_1 and h_2

$h_2 = h_f + x h_{fg}$

entropy $= s_2 - s_1$

superheated

$p_1 = 1\,MPa$, $T_2 = 200°C$ table A-4 → $h_1 = 2827.9\ \frac{kJ}{kg}$, $s_1 = 6.6940\ \frac{kJ}{kg}.K$

saturated

$T_2 = 40°C$, table A-2 → $h_{f2} = 167.57\ \frac{kJ}{kg}$, $h_{fg2} = 2406.7\ \frac{kJ}{kg}$, $h_2 = h_{f2} + x_2 h_{fg2}$

$$h_2 = 167.57 + 0.83(2406.7) = 2165.13\ \frac{kJ}{kg}$$

work ↑ power by turbine $= (+)$

a) $\cancel{\dot Q}^{0} - \dot W = \dot m\left[h_f - h_i + g(\cancel{Z_f - Z_i})^{0} + \frac{1}{2}(\cancel{V_f^2 - V_i^2})^{0}\right]$

$$\frac{\dot W_{cv}}{\dot m} = h_1 - h_2 = 2827.9 - 2165.13 = \boxed{662.769\ \frac{kJ}{kg}}$$

$s_{f2} = 0.5725\ \frac{kJ}{kg.k}$, $s_{g2} = 8.2570\ \frac{kJ}{kg.k}$

$$s_2 = s_{f2} + x_2(s_{g2} - s_{f2})$$

$$s_2 = 6.5725 + 0.83(8.2570 - 0.5725) = \boxed{6.951\ \frac{kJ}{kg.k}}$$

b) $\Delta S = S_2 - S_1$

$$\Delta S = 6.9506 - 6.6940 = \boxed{0.2566\ \frac{kJ}{kg.K}}$$

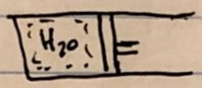
2 2 **15 / 15**

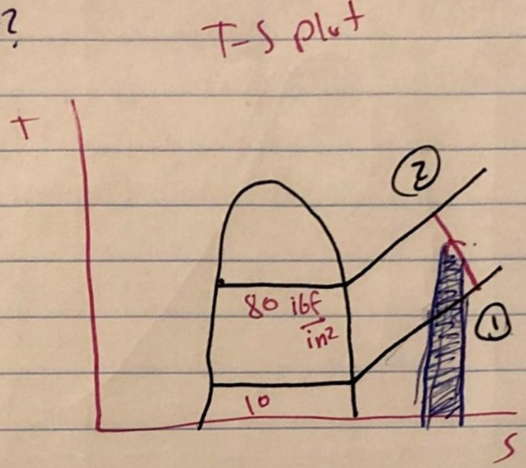
✓ **- 0 pts** Correct

- 2 pts Delta S is positive

$\boxed{3}$ $W = ?$ $Q = ?$

T-S plot





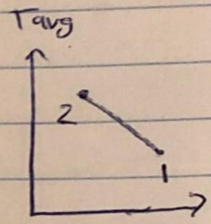


Table A-4E

$P_1 = 10\ lbf/in^2 \quad T_1 = 500°F$

$h_1 = 1287.7\ Btu/lb$

$u_1 = 1182.2\ Btu/lb$

$s_1 = 1.769\ Btu/lb \cdot °R$

$P_2 = 80\ lbf/in^2 \quad T_2 = 800°F$

$h_2 = 1430.4\ Btu/lb$

$u_2 = 1292.4\ Btu/lb$

$s_2 = 1.87\ Btu/lb\,°R$

heat transfer ?

$\boxed{\Delta T \text{ and } \Delta S}$ $\quad Q = \int^2 T\,ds = T_{avg}(s_2 - s_1)$

$Q_{12} = \int_1^2 T\,ds \Rightarrow \frac{Q_{12}}{m} = \int_1^2 T\,ds$

$\frac{Q_{12}}{m} = T_{avg}(s_2 - s_1) \Rightarrow \frac{Q_{12}}{m} = \frac{1}{2}(T_1 + T_2)(s_2 - s_1)$

$\frac{Q_{12}}{m} = \frac{1}{2}\left[(500+460)+(800+460)\right](1.87 - 1.769)$

$\frac{Q_{12}}{m} = \boxed{-109.89\ \frac{Btu}{lb}}$ in the process

Work?

internal energy $\quad u_2 - u_1 = \frac{Q_{12}}{m} - \frac{W_{12}}{m}$ $\qquad W = Q - (u_2 - u_1)$

$u_1 = 1182.2\ Btu/lb,\ u_2 = 1292.4\ Btu/lb,\ \frac{Q}{m} = -109.89$

$u_2 - u_1 = \frac{Q}{m} - \frac{W}{m} \Rightarrow 1292.4 - 1182.2 = -109.89 - \frac{W}{m}$

$\boxed{\frac{W}{m} = -220.09\ Btu/lb}$ by the system

Scanned with CamScanner

3 3 **15 / 15**

✓ **- 0 pts** Correct

- 7.5 pts W = ?

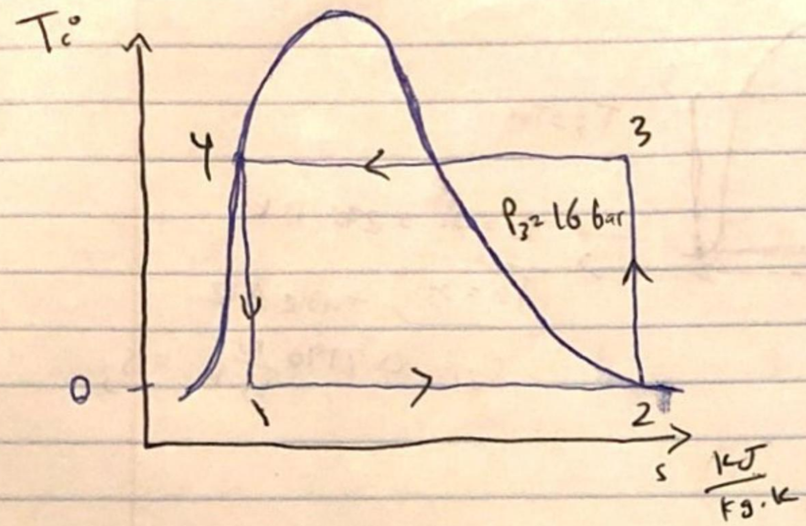
- 2 pts Work is negative

- 2 pts Q is negative

- 3.5 pts Wrong value for s1. s1 = 1.9690 Btu/(lb*R)

- 2 pts Wrong conversion from oF to oR

$\boxed{4}$



$$B = \frac{T_C}{T_H - T_C}$$

$T_c = 273.15\,K$ (0°c).

Also, $T_H = T_3$ To detemine T_3 apply $S_3 = S_2$

at 3: $S_2 = 0.9190 \quad \frac{kJ}{kg \cdot K}$

(Table A-10)

with $P_3 = 16\,bar$, Table A-12 gives $T_3 = 63.12\,°c$
or $T_3 = 336.27\,K$.

$$\therefore B = \frac{273.15}{336.27 - 273.15} = \boxed{4.33}$$

4 4 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** Th = 336 K (63 oC)
- **- 5 pts** CoP = Tc/(Th-Tc)
- **- 15 pts** No answer

5) $\rightarrow$ [generator] $\rightarrow$ $I = 230\,V$, $V = 13000\,V$

$\omega = 1800\,rpm$ $\quad$ $T_o = 298\,K$

$\tau = 16700\,N.m$

a) $\frac{dE}{dt}^{\to 0} = \dot{Q} - \dot{W} \rightarrow = \dot{Q} - (\dot{W}_{shaft} + \dot{W}_{electric})$

$T_b = \frac{-157.88 \times 10^3}{-110(32)} + 298 = \boxed{342.85\ K}$

$\dot{W}_{shaft} = -16700\,N.m \left(1800 \frac{rev}{min}\right)\left(2\pi \frac{rad}{rev}\right)\left(\frac{1 min}{60 sec}\right)\left(\frac{1 kJ}{1000}\right)$

$\dot{W}_{shaft} = -3147.88$

$\dot{Q} = \underbrace{\dot{W}_{shaft}}_{=\tau\omega} + \underbrace{\dot{W}_{electric}}_{iV}$

$iV = 230(13000) = 2990000\,W \frac{1 kW}{1000 W} = 2990\,kW$

$\dot{Q} = -3147.88 + 2990 = -157.88\,kW$

$\dot{Q} = -hA(T_b - T_o)$

$= -157.88 \times 10^3 = -110(32)(T_b - 298)$

b) $\frac{dS}{dt}^{0} = \sum_i \frac{\dot{Q}_j}{T_i} + \dot{\sigma}_{cv}$

rate of entropy

$0 = \frac{\dot{Q}}{T_b} + \dot{\sigma}_{cv} \rightarrow \dot{\sigma}_{cv} = -\left(\frac{-157.88}{342.85}\right) \Rightarrow \boxed{\dot{\sigma}_{cv} = 0.4605\ \frac{kW}{K}}$

replace T_b to T_o

c) $\frac{dS}{dt}^{0} = \sum_i \frac{\dot{Q}_i}{T_j} + \dot{\sigma}_{cv} \rightarrow 0 = \frac{\dot{Q}}{T_o} + \dot{\sigma}_{cv} \rightarrow \dot{\sigma}_{cv} = -\left(\frac{-157.88}{298}\right)$

$\boxed{\dot{\sigma}_{cv} = 6.5298\ \frac{kW}{K}}$

5 5 **20 / 20**

✓ **- 0 pts** Correct

- 2 pts c) the rate of entropy production is positive

- 3 pts a) Welectric = 2,990 kW. Q = -158 kW.

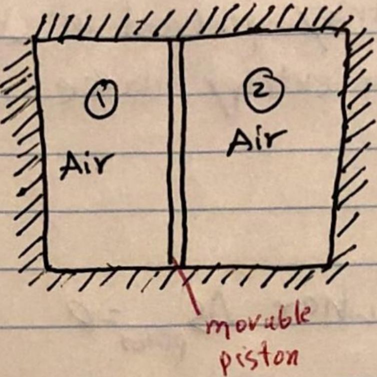
- 6 pts c) rate of entropy production = -Q/To = 0.53 kW/K

⑥

$V_1 = 1.0\,m^3$

$T_1 = \cancel{4}00\,k$

$P_1 = 3\,bar$



$V_2 = \cancel{1}00\,m^3$

$T_2 = \cancel{4}00\,k$

$P_2 = \cancel{1}.5\,bar$

Energy balance

a)

$\Delta U = Q - W = 0$

$\Delta U = \Delta U_1 + \Delta U_2 + \Delta U_{piston}$ $\quad\quad \Delta U_{piston} = 0$

$\Delta U = m_1(u_f - u_i) + m(u_{2f} - u_{2i}) = 0$

$u_{1i} = u_{2i} = u(400\,k)$ since the initial temp of each quanity of air is 400k

$u_{1f} = u_{2\,final} = u(T_f)$

$(m_1 + m_2)u(T_{initial}) = (m_1 + m_2)\,u(T_f)$

~~$T_f = 400k$ as expected.~~ $\quad T_F = 400\,k$ as expected.

b)

$P_{1final} = P_{2final}$ $\quad m = m_1 + m_2$ $\quad V = V_1 + V_2 = 3\,m^3$

$PV = nRT \Rightarrow PV = mRT$

$P_f = \dfrac{mRT}{V} = \dfrac{(m_1+m_2)RT}{V}$

m_1 and $m_2 = \left(\dfrac{PV}{RT}\right)$

$P_f = \dfrac{mRT_f}{V} = \dfrac{(m_1+m_2)RT_f}{V}$

where $m_1 = \dfrac{P_{1i}V_{1i}}{RT_i} = \dfrac{P_{1i}(\frac{V}{2})}{RT_f}$

$m_2 = \dfrac{P_{2i}(\frac{V}{2})}{RT_f}$, $P_f = \dfrac{P_{1i}(\frac{V}{2}) + P_{2i}(\frac{V}{2})}{V} = \dfrac{P_{1i} + P_{2i}}{2} = \dfrac{3\,bar + 1.5\,bar}{2}$

$P_f = P_{1f} = P_{2f} = \cancel{3\,bar}\ \boxed{2.25\,bar}$

(c) To determine the amount of entropy produced in $\frac{kJ}{kg}$ during the adiabatic process, an entropy balance reduces as follows:

$$\Delta S = \sigma$$

$$\sigma = \Delta S_1 + \Delta S_2 + \Delta S_{piston} \quad \text{where } \Delta S_{piston} = 0$$

$$\sigma = m_1(s_{1f} - s_{1i}) + m_2(s_{2F} - s_{2i})$$

$$\sigma = m_1\left(s^\circ_{1f} - s^\circ_i - R\ln\frac{P_{1\,final}}{P_{1\,initial}}\right) + m_2\left(s^\circ_{2F} - s^\circ_{2\,initial} - R\ln\frac{P_{2f}}{P_{2i}}\right)$$

initial and final temp = (300 K) so the s° terms cancel and we get

$$\sigma = m_1\left(-R\ln\frac{P_{1F}}{P_{1i}}\right) + m_2\left(-R\ln\frac{P_{2F}}{P_{2i}}\right)$$

sub eq ① and ②

$$\sigma = -\frac{V}{2T_F}\left[P_{1i}\left(\ln\frac{P_{1F}}{P_{1i}}\right) + P_{2i}\left(\ln\frac{P_{2F}}{P_{2\,in}}\right)\right]$$

$$\left[3\,bar\,\ln\left(\frac{2.25}{3}\right) + 1.5\,bar\left(\frac{2.25}{1.5}\right)\right]$$

$$\sigma = \frac{-2\,m^3}{2(400\,K)}$$

$$\cancel{3.186\times10^{-4}} \times \left|\frac{100\,\frac{kN}{m^2}}{1\,bar}\right| \cdot \left|\frac{1\,kJ}{1\,kN\cdot m}\right| = 0.0637\,\frac{kJ}{K}$$

6 6 **20 / 20**

✓ **- 0 pts** correct

- 5 pts (1) definition of delta Sp; (2) why delta Sp =0; (3) why s1f, s1i, s2f, s2i were not included for the following calclation?

- 3 pts Its 400K based on the given question

- 1 pts unit of K not degree K

- 5 pts The delta S equation for idea gas was incorrect

- 6 pts final P =?

- 2 pts entropy produce calculation incorrect, P =400 noT 300K

- 6 pts entropy produce=?

- 20 pts no submission

- 5 pts delta S calculation incorrect

- 1 pts Unit for T2

- 3 pts calculation of m1 & m2=? Final P incorrect

- 3 pts (1) why the delta piston was mentioned at the beginning and was not included in the following calculation? need the explanation for each step if it was neglected